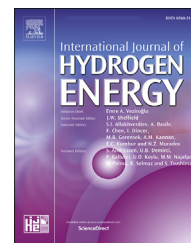


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An improved CFD model for vented deflagration simulations – Analysis of a medium-scale hydrogen experiment

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ABSTRACT

Hydrogen is a promising alternative fuel which is expected to gain a significant portion of the energy market in the near future. However, it is a flammable gas with significant safety concerns as it can easily cause deflagrations or even detonations. The aim of this work is twofold: firstly, to derive a new CFD model for vented deflagration simulations improving an existing one and secondly, to better understand the physics of the phenomenon. A recent experiment conducted in Karlsruhe Institute of Technology is used for that purpose. The new model improves substantially the predictions representing successfully the experiment. The analysis indicates that turbulence generated outside the vent is responsible for the violence of the external explosion which leads to the sudden pressure increase. Rayleigh-Taylor instability on the other hand does not seem to contribute to burning velocity increase. From the analysis that is made combustion models are formulated highlighting the important components for a successful CFD deflagration model.

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Introduction

Within recent years hydrogen has gained a lot of attention due to the prospective of being used as a clean energy carrier contributing to greenhouse gas emissions reduction. However it is known that hydrogen is a highly flammable gas that can result in deflagrations or even detonations in the case of an accident. Consequently, hydrogen safety [1] is a very active research area helping hydrogen to be established in the energy market.

Numerical models have been used from decades in order to predict deflagration hazards for risk assessments or for design purposes (e.g. vent design [2]). Even though analytic or semi-

empirical models have been proved useful for quick calculations, their applicability is limited to simple cases. Computational Fluid Dynamics (CFD) can provide a more accurate tool in order to predict the overpressure that is generated from a deflagration by solving the differential equations which describe the fluid dynamics. Its capability of taking into account the exact geometry of the problem is a major asset.

However, CFD modelling of deflagration is a challenge. Rate of combustion needs to be estimated correctly in order to have an accurate prediction of flame speed and overpressure. This is a difficult task due to various phenomena that influence deflagrations such as flame instabilities and turbulence. In practical applications, combustion occurs in scales that are

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much smaller than the scale of the geometry. As a result, models are needed in order to simulate the unresolved phenomena. Unfortunately, there is no model that has been proved capable of simulating with accuracy all the relevant deflagration cases. As a result many CFD models have appeared in the literature and many benchmark exercises have been conducted in order to compare and evaluate them (e.g. Ref. [3–15]).

Vents are an effective way to mitigate the hazards of an accidental explosion. Vented deflagrations are especially interesting and have been studied extensively (e.g. Ref. [16–21]) due to the complexity of the phenomenon. Many pressure peaks can occur as a result of different phenomena. The external explosion effect, i.e. the increase of the pressure inside the enclosure due to the combustion of the unburned mixture outside the vent, is very often the dominant overpressure peak (e.g. Ref. [22–24]). In this work, we study the modelling of vented deflagrations focusing on the external explosion. We do not deal with flame-acoustic instabilities which usually occur at the latest stages of the explosion (e.g. Ref. [17,20,25]).

The “Multi-phenomena turbulent burning velocity” combustion model [26,27] will be used initially. This model has been evaluated successfully against a variety of cases (e.g. Ref. [3,5,7,26–29]). A drawback of the model is the fact that it significantly underpredicts the external explosion phenomenon [30–32]. In an effort to improve the model, Keenan et al. (2014) [31] incorporated a submodel for the Rayleigh-Taylor instability which was believed to be responsible for the intense violence of the external explosion. The model increased the intensity of the external explosion leading to better prediction of the first overpressure peak. The purpose of the current work is to evaluate the initial “Multi-phenomena turbulent burning velocity” model [26,27] against a new vented deflagration experiment and investigate whether or not modelling of the Rayleigh-Taylor instability is necessary in order to predict the external explosion. We show that if improvements in the initial model are made, the external explosion can accurately be predicted without the need of a Rayleigh-Taylor instability submodel. Thorough analysis of the experiment is made discussing the physics of the phenomenon.

The vented deflagration experiment conducted in Karlsruhe Institute of Technology (KIT) [25] was chosen to be simulated. This experiment is very suitable for model validation for a variety of reasons. Firstly, it is a contemporary experiment with raw experimental data available. Secondly, its credibility is high because data for experiment repeatability exists demonstrating the experimental uncertainty. Therefore, the evaluation of a numerical model can be made more effectively and reliably. Thirdly, the external (outside the vent) pressure is also measured during the deflagration and thus the external explosion effect can be evaluated. Fourthly, the effect of vent cover, which is difficult to be modelled accurately, is minor because it was removed prior to ignition. Finally, the deflagration was recorded using the Background Oriented Schlieren (BOS) technique [33]. Consequently, clear photos of the flame are available (which is something rare in hydrogen deflagrations).

This experiment has been modelled again in the framework of the HyIndoor [34] and SUSANA projects [35]. In HyIndoor project four different CFD codes were evaluated by comparing the overpressure curves. Even though the maximum overpressure inside the enclosure was predicted accurately enough by all codes, the overall qualitative agreement of the overpressure time-series was not good. In SUSANA project the experiment was modelled [36] using the COM3D code. The maximum value of the overpressure agreed well with the experiment, but nonetheless, a distinctive overpressure drop not appearing in the experiment was also predicted. Moreover, the model overestimated the flame speed. Veyzenka and Jallais [9] simulated also the same experiment in a validation study of FLACS program. Only the maximum overpressure was presented which agreed well with the experimental one. In this paper, the overpressure time-series is tried to be predicted with accuracy until the time of the external explosion, predicting all the important qualitative features.

The paper is organized as follows: In Section **Experiment overview** the experiment is described and the experimental overpressure time histories are presented and discussed. In Section **Modelling approach** the main equations and the initial combustion model that is used are presented. In Section **Results of the initial combustion model** the results of the initial deflagration model are demonstrated. In Section **New model: derivation and results** the new deflagration model is presented and evaluated in two steps: First, a key modification of the initial combustion model is made (Section **Main modification of the combustion model**). Second, a further modification of the combustion model along with a modified turbulence model is utilized composing the new deflagration model (Section **New model**). In Section **Analysis of the experiment using the new model** the analysis of the experiment is presented using the new model, which achieved the best agreement with the experiment. Finally, **Discussion** is made and **Conclusions** are summarized.

Experiment overview

The dimensions of the enclosure are 0.98 m × 0.96 m × 1.00 m (length, width and height: x, y, z coordinates). The vent is square with size of 0.5 m × 0.5 m positioned at the end of the enclosure length. The modelled geometry along with a photo from the experimental facility is shown in Fig. 1. The vented enclosure is positioned inside a room of 5.5 m × 8.5 m × 3.4 m. The walls of the room are covered with damping material and special corrugated metal sheets to absorb shock waves. The distance of the right room wall from the enclosure closest side is 1.94 m, of the left wall 2.6 m, of the back wall 1.95 m and of the ground 0.62 m. Outside the vent a beam structure exists which holds the measurements devices. This structure is included in the simulations (Fig. 1). The distance of the horizontal beam edge from the vent is set equal to 0.15 m.

In the experiments, a thin plastic film was used as hermetic cover for the vent during the filling procedure. It was removed prior to ignition to avoid any influence on the results. In the tests that we examine in this work, the film was broken

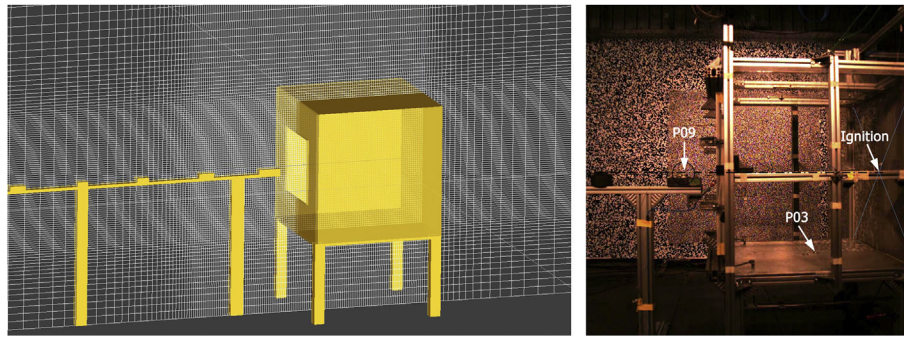


Fig. 1 – Left: Modelled geometry and XZ plane of the computational domain (Grid 3). Right: Photo from the experiment (sensors and ignition point are indicated).

by a set of two cutting wires that were activated (heated) electrically. After the destruction of the film the mixture was ignited by a spark generated between two electrodes in a point inside the enclosure [34].

The pressure was measured at several positions inside and outside the enclosure. Two positions will be used here to compare with the experiment, pressure P03 located inside the enclosure at the centre of its bottom side and pressure P09 located 0.24 m outside the vent at the centreline (Fig. 1).

A large number of experiments were conducted. In this work, we study the experiment of homogenous hydrogen-air mixture of 18% v/v (volumetric) hydrogen concentration and ignition at the centre of the back-wall. Four experiments were conducted under these conditions in order to test repeatability. The code numbers of these experiments were HIWP3-028, HIWP3-029, HIWP3-030, HIWP3-031. In this paper, for simplification, these names are referred as 028, 029, 030 and 031 respectively. In order to achieve homogenous mixture inside the enclosure, the hydrogen-air mixture with the desired concentration was injected at the top of the enclosure displacing the atmospheric air which escaped the enclosure from the bottom of the front plate. When the hydrogen concentration at the escaped mixture was equal to the injected one the process was stopped. In addition to that, the concentration in other representative points in the enclosure was measured in order to ensure homogeneity. More details about the experiment can be found in Ref. [25,34].

In Fig. 2 (top) the internal and external overpressure time series are presented for each repetition of the experiment. We observe that the results of the experiments 029, 030, 031 are very similar with each other, at least until the time of the main pressure peak which is the area of interest. The maximum (filtered) overpressure for those experiments lies between 12.05 and 13.17 kPa for the internal one and between 7.56 and 8.69 kPa for the external one.

Experiment 028 is the only one which differs noticeable from the rest. The differences, however, is only quantitative because all overpressure curves exhibit similar behaviour: As the flame propagates inside the enclosure the internal pressure gradually increases until the flame exits the vent at approximately 0.078 s for experiments 029–031 and 0.086 s for experiment 028 (photos of the flame at these times for experiments 030 and 028 are presented in Fig. 2). The internal pressure remains steady for a small time interval in all

experiments and then suddenly increases due to the external explosion, i.e. the combustion of the unburned fuel that has been pushed outside the vent. The sudden increase of the external pressure occurs at about 0.082 s and 0.089 s for experiment 028 and 030 respectively (photos of the flame at these times are presented in Fig. 2). The reason for the quantitative differences of experiment 028 is not clear to the experimentalists. The propagation speed inside the enclosure is slower compared to the speed in the other experiments leading to a lower rate of pressure rise (mainly while the flame propagates inside the enclosure) and lower maximum overpressure peaks.

In the bottom of Fig. 2 the internal and external overpressure curves are presented together for each experiment. We observe that the external explosion effect appears in all experiments. The overpressure inside the enclosure increases rapidly reaching the maximum value after the sudden increase of the external one. The external explosion effect is confirmed in the examination of the flame front position from the video recordings [25,34]. In these videos it is shown that unburned mixture is pushed outside the enclosure during the propagation of the flame inside the enclosure from the back wall to the vent. Once the flame has exited the vent, this mixture is ignited leading to the external overpressure peak. We should note that, as it is clear from the flame photos in Fig. 2, the sudden increase of the external pressure occurs at the later stage of the external explosion and not immediately after the exit of the flame from the vent.

Modelling approach

The one step irreversible reaction equation for hydrogen combustion is used. The equations that are solved are the continuity equation, the Navier-Stokes equations, the conservation equation for the total enthalpy and the conservation equations of the mass fraction of hydrogen, oxygen and water. The multi-component mixture is assumed to be in thermodynamic equilibrium. The ideal gas equation of state is used. The conservation equations are ensemble-averaged in the case of the URANS (Unsteady Reynolds averaged Navier-Stokes equations) approach for turbulence modelling or implicit-filtered in the case of LES (Large Eddy Simulation). In this work, the $k-\epsilon$ model is used in the URANS case whereas the

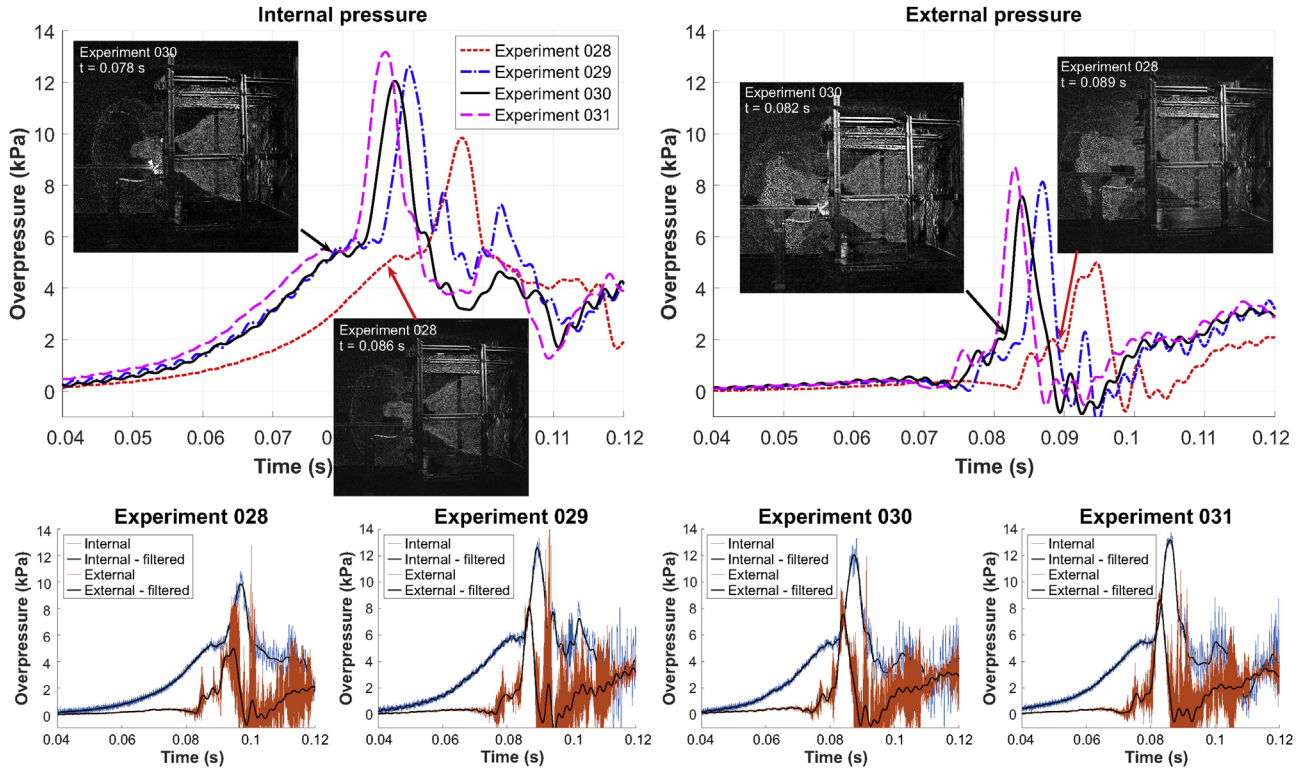


Fig. 2 – Top: Low pass filtered (400 Hz) internal and external pressure for the four repetitions of the experiment. Bottom: Internal and external pressure (filtered and unfiltered) in the same plot for each repetition of the experiment – the external explosion effect is depicted.

RNG method [37,38] for LES. The reaction rate for hydrogen which appears in its conservation equation is modelled as [39]:

$$\bar{R}_f = \rho_u S_T |\nabla q_f| \quad (1)$$

where ρ_u is the density in the unburned region, S_T the turbulent burning velocity and q_f the fuel mass fraction. The reaction rates for oxygen and water can be easily estimated from the above rate through the reaction stoichiometry [40]. ρ_u is estimated using the assumption of adiabatic compression (which stands with accuracy [41]) whereas the turbulent burning velocity estimation is presented in the subsequent sections.

Following the Best Practise Guidelines [42] many sensitivity studies were conducted in order to increase the fidelity of the results, as it is shown throughout the text. **The ADREA-HF CFD code is used [43]. ADREA-HF utilizes the finite volume method on a staggered Cartesian grid to discretize the conservation equations. The geometry is reproduced with the use of porosities, which makes possible the correct representation of any solid surface on a structured mesh. In the present work, the grid is aligned with all the obstacles and, thus, the porosities method is not utilized.** The pressure and velocity equations are decoupled using a modification of the SIMPLER algorithm [44]. For the discretization of the convective terms the bounded second order upwind method is used [45]. The implementation is carried out using a deferred-correction approach via the source term [46]. For time advancement, the fully implicit first order accurate scheme was chosen. The Crank-Nicolson scheme was also tested in LES, exhibiting

minor changes in the results. Time step is automatically adapted according to desired Courant–Friedrichs–Lewy (CFL) number which was set equal to 0.2. Smaller CFL numbers were examined and no differences in the results were found.

The computational domain includes the interior of the room where the vented enclosure resides. No-slip boundary conditions were used in all solid boundaries and standard wall functions were applied. The fact that the surrounding room was considered totally sealed (whereas in the experiment gaps existed or small vents opened during the deflagration to vent the pressure) and no shock waves absorption at the walls was considered, has a negligible influence on the maximum overpressure. This was inferred by a simulation conducted with open ‘non-reflecting’ type of boundary conditions (in all domain sides except bottom) and bigger domain size (using the model of Section **New model**). The maximum overpressure in that simulation was decreased by only 3%.

A stagnant initial velocity field was assumed. Very small initial values of k and ϵ in the case of k - ϵ turbulence model were chosen, equal to $1.0\text{e-}06 \text{ m}^2/\text{s}^2$ and $1.0\text{e-}06 \text{ m}^2/\text{s}^3$ respectively, which basically correspond to an initial zero turbulent field. Inside the enclosure the initial mass fraction of hydrogen and oxygen was set equal 0.01511 and 0.2294, respectively. Outside this area the mass fraction of oxygen was set equal to 0.232. Nitrogen was the inert species. Ignition was modelled by fixing the reaction rate in the cell at the point of ignition, in order the initial amount of fuel to be burned at a determined time [29]. This time was set equal to 2 ms. It was approximately estimated by the formula $0.5dx/(E \cdot S_{L0})$ where

dx is the size of the cell at ignition point, $E = 5.2$ the expansion coefficient and $S_{L0} = 0.64$ m/s the laminar burning velocity at the initial pressure.

Three computational grids were used in order to assess grid independence, a coarse one with approximately 714,000 cells, a medium one with 1,539,000 cells and a fine one with 2,726,000 cells. The results exhibit small grid dependency as it is presented in Section [New model](#). The fine grid was used in most simulations except otherwise mentioned. The control volumes size was approximately uniform inside the enclosure and outside until the distance of 0.8 m from the vent ([Fig. 1](#)) and equal to 0.03, 0.02 and 0.015 m for the coarse, medium and fine grid respectively. The expansion ratio outside that uniform region was set equal to 1.08. The modelled geometry is presented in [Fig. 1](#). The vent thickness was discretised using 1 cell.

For the simulations a modern processor (Intel Xeon E5-2690 v3, 2.6 GHz) was used. The simulations run in parallel in four of its cores. The computational time was approximately equal to 19, 61 and 118 h for the coarse, medium and fine grid for 0.12 s simulation time using the final model ([Section New model](#)).

Initial combustion model

The “Multi-phenomena turbulent burning velocity” model [\[26,27\]](#) developed in the context of LES at the University of Ulster is used first. This model (and its variations) has exhibited very good predictive capabilities in a variety of cases (e.g. Ref. [\[3,5,7,26–29\]](#)). This model was incorporated into the ADREA-HF code confirming its good performance [\[40,47\]](#). It was also evaluated in combination with URANS approaches for turbulence instead of LES preserving the good behaviour [\[40,47\]](#). A drawback of this model seems to be the incapability to predict the sudden pressure increase in vented deflagrations due to the external explosion. Thus extensions of the combustion model have been proposed to resolve this issue which either simply increase the reaction rate outside the vent by a factor (between 1 and 2) [\[30,32,48\]](#) or incorporates additional complex sub-model to account for Rayleigh-Taylor instability [\[31\]](#). The “Multi-phenomena turbulent burning velocity” model is initially used in the present work without any modification for the external explosion in order to evaluate its performance against the KIT vented deflagration experiment. This model is described in the following paragraphs.

The model considers all the main mechanisms that affect the turbulent burning velocity in hydrogen deflagrations. It takes into account the: laminar burning velocity, turbulence existing in front of the flame front, turbulence that is generated by the flame front itself, non-equidiffusive effects and fractal structure of the flame front surface. The effect of the fractal structure is important in large scale deflagrations only and can be omitted from the model in the present study as it does not influence the results. Turbulent burning velocity is estimated from the relation:

$$S_T = \Xi_k \cdot \Xi_{lp} \cdot S_L \cdot \exp\left(\frac{u'}{S_T}\right)^2 \quad (2)$$

The above relation is based on Yakhot's equation [\[49\]](#) for premixed turbulent combustion. The Ξ factors are not

included in the original form of Yakhot's equation and have been incorporated, in order account for the turbulence that is generated by the flame front itself (Ξ_k) and non-equidiffusive effects (Ξ_{lp}). Thus, the main idea of the model is to substitute the laminar burning velocity S_L with the subgrid scale burning velocity $S_L^{sgs} = \Xi_k \cdot \Xi_{lp} \cdot S_L$ to account for the subgrid scale phenomena which enhance the burning velocity and cannot be resolved on practical computational grid resolutions. Eq. (2) is solved iteratively, using Newton-Raphson and bisection method.

Ξ_k accounts for turbulence that is generated by the flame front itself. This phenomenon is driven by the hydrodynamic instability, and as a result Ξ_k depends on the expansion coefficient E (i.e. ratio of densities in the unburned and burned area which is equal to 5.2 for the mixture of the present study). Hydrodynamic instability needs some distance, R_0 , in order to be fully developed, which is taken here equal to 1 m (the values that are usually used are 1 or 1.2 m based on the work of Gostintsev et al. [\[50\]](#)). In order to account for the transient stage, Ξ_k is estimated from the relation $\Xi_k = 1 + (\psi \cdot \Xi_k^{\max} - 1) \cdot [1 - \exp(-R/R_0)]$ where R is the distance from the ignition point and $\Xi_k^{\max} = (E - 1)/\sqrt{3}$ [\[51\]](#). ψ is a model constant varying between 0 and 1 which represents the level at which the maximum value $\Xi_k^{\max}$ is reached. This constant is suggested to be equal to 0.5 for near stoichiometric mixtures and equal to 1 for lean hydrogen-air mixtures [\[27\]](#). Thus, the value of 1.0 is used in this study.

Ξ_{lp} accounts for non-equidiffusive effects. Non-equidiffusive effects (i.e. nonunity Lewis number and preferential diffusion effects [\[52,53\]](#)) can influence the combustion affecting both laminar and turbulent flames [\[54\]](#). In lean hydrogen mixtures, non-equidiffusive effects make the flame unstable wrinkling its surface and increasing the combustion rate [\[53,55\]](#). This instability, which is often called diffusional-thermal, develops faster than the hydrodynamic one and thus it influences the deflagration process at the very early stages [\[56\]](#). Ξ_{lp} increases linearly from the value of one to a constant maximum value at a given distance where the instability is considered to be fully developed. This distance is proposed to be set equal to $R_0/2$ [\[26,27\]](#), i.e. 0.5 m in our case. In the present work, however, a smaller distance was adopted, equal to 0.1 m, because experimental data indicates that the instability is developed in much smaller distances (e.g. see Kim et al., 2015 [\[56\]](#)). Diffusional-thermal instability depends on Lewis number. Thus, the maximum value of Ξ_{lp} is a function of mixture composition. It is estimated [\[26\]](#) based on the leading point concept of Zimont and Lipatnikov work [\[57\]](#). For 18% hydrogen concentration it is approximately equal to 2. This is in agreement with the value of 1.96 which can be estimated from the experiments of Bauwens et al. (2011) [\[24\]](#). The value of 1.96 was used in this study. The fact that the effect of diffusional-thermal instability can be modelled by increasing the burning velocity by a constant factor is supported by the work of Bauwens et al. in Ref. [\[24,58\]](#).

u' accounts for turbulence existing in front of the flame front. u' is the rms (root mean square) of velocity fluctuations (subgrid part for LES). It is estimated from the relation $u' = \sqrt{2/3 \cdot k}$. In k - ϵ model k is directly computed through its transport equation. In LES, it is estimated from the formula

$\mu_{sgs}^2/(\rho^2 L_{sgs}^2)$ (which has been used again in relative applications, e.g. in Refs. [27,59,60]) where μ_{sgs} is the subgrid scale dynamic viscosity given by the RNG-LES method, $L_{sgs} = 0.157 \cdot V^{1/3}$ the subgrid length scale, ρ the density and V the computational cell volume.

Finally, the laminar burning velocity dependency on pressure is taken into account through the relation $S_L = S_{L0}(P/P_0)^\epsilon$, where P is the pressure and S_{L0} the laminar burning velocity at the initial pressure P_0 ($S_{L0} = 0.64$ m/s and $\epsilon = 0.66$ [5] in the 18% v/v mixture of the present study). The effect of pressure on laminar burning velocity is small in our case because overpressure is not high enough.

Results of the initial combustion model

In Fig. 3 the results of the initial combustion model are presented. We observe that the predictions underestimate significantly the maximum overpressure both inside and outside the enclosure. The choice of the method (LES or URANS k- ϵ) does not seem to have a great qualitative impact on the pressure curve. Both simulations have a good agreement with the experiment until the time when the flame exits the vent (approximately at 0.08 s). The overpressures stop increasing at this time, similarly to the experiment. However, the duration of the overpressures steadiness is much larger in the simulations than in the experiment. In the experiment the overpressure starts increasing much faster (due to the external explosion) compared to simulations in which a big delay is observed. Moreover, the pressure increase is significantly bigger in the experiment than in the simulations. The above facts lead to the conclusion that the external explosion is not correctly reproduced.

In the right part of Fig. 3 combustion model related variables are presented for LES and URANS k- ϵ method. These are the mean values of the variables at the flame front that exists only inside or outside the enclosure. According to the combustion model used, turbulent burning velocity S_T deviates from the subgrid scale burning velocity S_L^{sgs} only if $u' > 0$. As a result we observe that inside the enclosure the effect of turbulence on combustion (which is expressed through u') is minor for LES and almost zero for URANS k- ϵ . As a result flame

instabilities are responsible for the increase of the reaction rate inside the enclosure. Outside the enclosure the effect of turbulence gets bigger which leads to the deviation of S_T from S_L^{sgs} . However there is a significant qualitative difference between LES and k- ϵ . In LES the maximum value of u' , equal to 2 m/s, is almost immediately achieved. Then, u' steadily decreases. On the other hand, in k- ϵ case, u' steadily increases getting values bigger than 2 m/s. As a result turbulent burning velocity S_T also continually increases following u' trend. The values of S_L^{sgs} (green dotted line) are similar between LES and k- ϵ model because this variable is not linked to u' .

The greater values of S_T in the k- ϵ model case are depicted in the overpressure values. Even though the maximum internal overpressure is not significant higher compared to LES case, the ratio of the maximum overpressure to the overpressure at the time when the flame exits the vent is 20% higher. Moreover, the maximum external overpressure is 30% times higher which indicates the prediction of more violent external explosion.

More insight into models behaviour can be gained if we study u' contours. Fig. 4 displays the contours for LES (subgrid part) and k- ϵ along with the flame position at various times. The first time corresponds to the time when the flame exits the vent resulting in the overpressure stabilization, the second at the moment when the pressure outside the vent begins to increase and the third when the external pressure peak occurs. We observe that the contours differ both qualitatively and quantitatively. In LES case high values of u' occur mainly around the flame front whereas in the k- ϵ case high values are observed away from the flame front at the area of the recirculation formed due to the high speed flow coming out from the vent.

High values are also predicted at the area of the horizontal vent, mainly in the k- ϵ case. A reason for the high values in this area is the step in the geometry that is formed between the P09 sensor box and the horizontal beam (Fig. 1). However, high values are developed mainly after the flame passage and thus the reaction rate and overpressure are not influenced. This was confirmed in an additional simulation that was conducted eliminating the step by increasing the height of the horizontal beam: u' was decreased in the beam area without differences in overpressure.

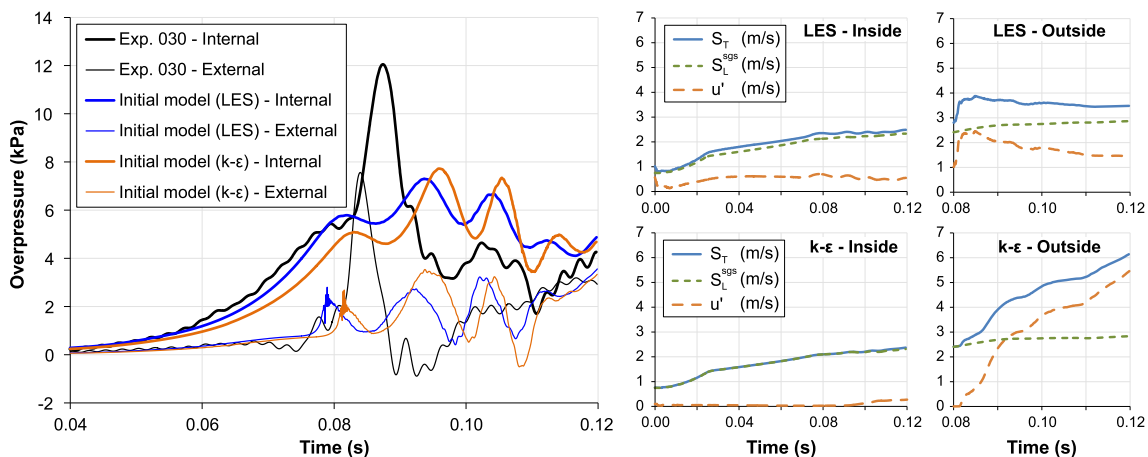


Fig. 3 – Results of the initial model. Left: Overpressure inside and outside the enclosure. Right: Average combustion related variables at the flame front inside or outside the enclosure for LES (top) and URANS k- ϵ method (bottom).

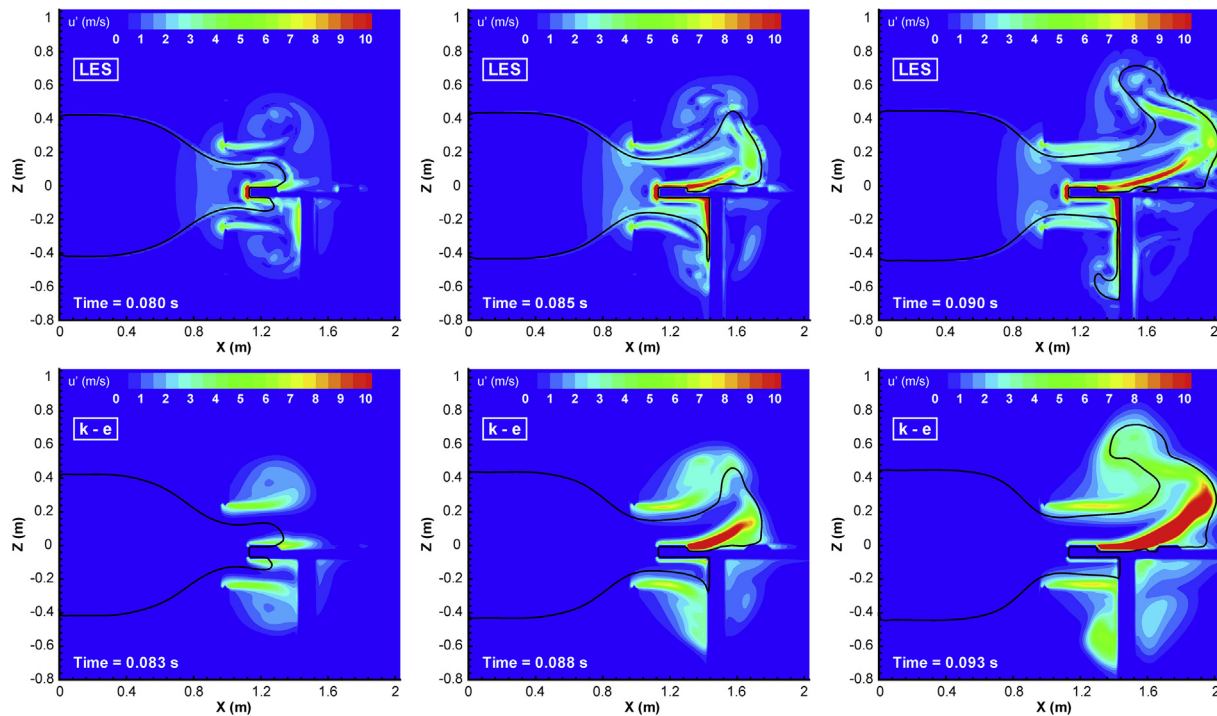


Fig. 4 – u' contours along with the flame front (black line) for LES (top) and URANS $k-\epsilon$ method (bottom) at various times at xz plane passing from the ignition point (simulations with the initial combustion model).

New model: derivation and results

Main modification of the combustion model

The comparison between the LES and the URANS $k-\epsilon$ simulations in the previous section revealed some interesting differences which put forward the following arguments in favour of URANS model compared to LES.

The first argument is the fact that URANS $k-\epsilon$ predicted 30% more violent external explosion compared to LES. High values of u' exist outside the vent at the area between the vent edges and the surroundings. The examination of the experimental photos (Fig. 2) shows that the sudden increase of the external pressure occurs at the moment when the flame enters this area. In this area vortices are expected to be formed and high turbulence to exist.

The second argument in favour of URANS $k-\epsilon$ model is that the flame front itself is not predicted to produce high values of u' , opposed to LES. This was also observed in the free unconfined hydrogen deflagration simulations in Ref. [47] and in the vented deflagrations in Ref. [8]. Colin et al. [61] have pointed out that in LES the dependency of subgrid dynamic viscosity (and thus u') to the strain rate can result in overestimation of u' at the flame front in the case of low turbulence due to thermal expansion, and thus result in overestimation of the turbulent burning velocity (in order to deal with this issue a new relation for the estimation u' was proposed, which is based on the vorticity field). Moreover, in modelling perspective, the turbulence generated from the flame front is desirable to be modelled separately (e.g through the $\mathcal{E}$ factors in the

examined model) and not to be enhanced with a contribution through the u' which is usually used to quantify the turbulence that exist only in front of the flame front. The issue of u' estimation is also stated by Poinot and Veynante [62]. They point out that in experiments is calculated in the unburned region whereas in most simulations is estimated inside the flame according to the local characteristics of the flow. In Fig. 4, u' seems to be more unaffected by the flame front in URANS $k-\epsilon$ model compared to LES.

The last argument in favour of URANS $k-\epsilon$ model is the fact that Yakhot's equation has not been developed specifically for the context of LES. As a result the filter scale is not considered in Yakhot's equation in contrast to combustion models that have been specifically developed for LES such as the models of Colin et al. [61] and Charlette et al. [63].

Consequently, in the followings the URANS method for turbulence is followed. The combustion model, which was initially developed in the context of LES, is improved and reformulated in order to fit better into the URANS approach.

An issue of the initial combustion model, which may hinder its predictive capabilities, is the substitution of the laminar burning velocity S_L with the subgrid scale burning velocity $S_L^{sgs} = \mathcal{E}_k \cdot \mathcal{E}_p \cdot S_L$. The modification that is examined in this section is not to make this substitution but to use the original equation of Yakhot. Yakhot's equation expresses a specific dependency between turbulence and burning velocity, using the laminar burning velocity as a parameter. If the laminar burning velocity is altered, then the predicted effect of turbulence is changed. Laminar burning velocity is an intrinsic characteristic of the flame. Turbulence tries to wrinkle the flame increasing the burning velocity whereas the

flame front tends to resist to those changes. The higher the laminar burning velocity the greater the resistance of the flame. Thus, by changing the laminar burning velocity we change the way that flame and turbulence interact.

However, as we saw in the previous sections, Ξ factors contribute significantly in the turbulent burning velocity and cannot be omitted. Thus, in the modified model that is proposed here, an ‘intermediate’ turbulent burning velocity S'_T is estimated using the original Yakhot's equation and then the Ξ factors are used to multiple S'_T accounting for the additional wrinkling of flame due to instabilities, that is:

$$S'_T = S_L \cdot \exp\left(\frac{u'}{S'_T}\right)^2 \quad (3)$$

$$S_T = \Xi_k \cdot \Xi_p \cdot S'_T \quad (4)$$

As a result, in our approach, Yakhot's equation remains unmodified accounting for the turbulence that exists in front of the flame front while the other phenomena (turbulence generated by the flame front itself and non-equidiffusive effects) are still considered. The above formulation implies that the accounted phenomena act on flame wrinkling at different scales and thus their effect accumulate (as the multiplication in eq. (4) represents). It can easily be proved (see Appendix) that the modified model predicts higher turbulent burning velocity values than the initial one. For example, for $u' = 5$ m/s, $S_L = 0.64$ m/s and $\Xi_k \cdot \Xi_p = 3$, the turbulent burning velocity increases by a factor of 2.2. The increase of the turbulent burning velocity compared to the initial model is something desirable in order to obtain a better agreement with the experiment.

In Fig. 5 (left) the overpressure time series are presented for the modified model and compared with the initial one (k- ϵ case) and with the experiment. It is obvious that the modification in the combustion model lead to significant improvement of pressure prediction. The overpressure curves are closer to the experimental one. The maximum overpressure both inside and outside the enclosure have increased as expected by the above analysis. However, the maximum values are still lower than the experiment. It is interesting to note

that the maximum values in the modified model are very close to the experiment 028 (the one which deviated from the rest predicting the lowest pressures, see Section 2) overpredicting the internal and external maximum overpressure by only 7%.

The modified model affects the results mainly after the flame has exited the vent. This is normal because the modified model predicts different values of turbulent burning velocity only in the areas in which u' is not negligible. This is demonstrated also by the plots in the right part of Fig. 5. We observe that inside the enclosure the combustion related variables have the same values as in the initial model. On the other hand, outside the enclosure, the modified model predicts much higher values of turbulent burning velocity even though the values of u' and S_L^{sgs} are similar (at least until 0.10 s when external explosion is completed).

New model

The modified model presented in the previous section led to significant improvement of the predictions. Even though the qualitative agreement with the experiment is rather good, the maximum overpressure peak is still underpredicted. Thus, the deflagration model presented in the previous paragraph is further modified. Two changes are made. **The first change concerns the combustion model and the second one the turbulence model.**

Regarding the combustion model, the equation of Schmid et al. (1998) [64] is used instead of Yakhot's equation in order to account for the turbulence that exists in front of the flame front. This relation is more recent than the one proposed by Yakhot. Moreover, it takes into account not only u' and laminar flame speed but also the Damköhler number (Da):

$$\frac{S_T}{S_L} = 1 + \frac{u'}{S_L} (1 + Da^{-2})^{-1/4} \quad (5)$$

The dependency on Damköhler number makes this model capable of predicting the turbulent burning velocity in the whole range of combustion regimes [64]. This equation predicts well experiments with propane-air and hydrogen-air mixtures [64] and it has been used again in the past in order

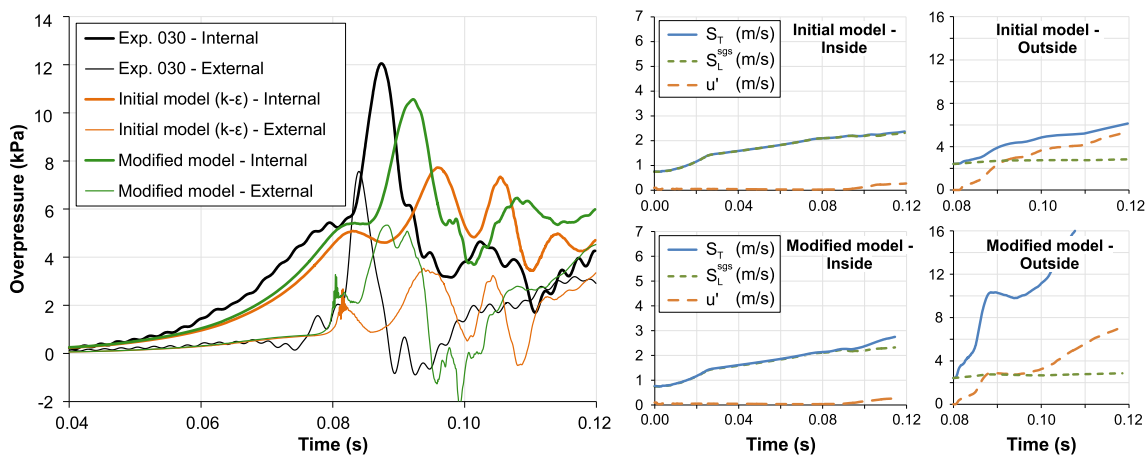


Fig. 5 – Results of the modified model and comparison with the initial one. Left: Overpressure inside and outside the enclosure. Right: Average combustion related variables at the flame front inside or outside the enclosure for the initial model (top) and the modified one (bottom).

to model vapour cloud explosions, e.g. in Ref. [65,66]. The Damköhler number is defined as the ratio of turbulence to chemical time scale. As turbulence time scale we use the ratio of k/ϵ whereas the chemical time scale is estimated as the ratio of the laminar flame thickness to the laminar burning velocity. The laminar flame thickness was taken equal to 0.37 mm [56] and as a result the chemical time scale is estimated equal to 5.780×10^{-4} s ($S_{L0} = 0.64$ m/s).

The substitution of equation (3) with equation (5) led to the increase of the maximum overpressure. This increase however was too high. The maximum overpressure inside the enclosure was equal to 18.4 kPa and outside equal to 13 kPa (the medium grid was used in this simulation). This can be justified by comparing Yakhot's and Schmid's equations. Fig. 6 presents the plots of these equations. We observe that as the Damköhler number increases, Schmid's equation predicts greater values of S_T/S_L . However this increase is very small for high Damköhler numbers, for example the lines of $Da = 10$ and $Da = 100$ are almost identical. We observe also that Schmid's relation predicts higher values of S_T/S_L compared to Yakhot's relation for Damköhler numbers greater than about 0.4. Damköhler number is higher (equal to 10 approximately) than this value at the area of recirculation outside the vent and as a result the turbulent burning velocity increases when Schmid's equation is used.

Thus, a modification in turbulence modelling was also applied. Particularly, the Kato and Launder modification of the $k-\epsilon$ model was used [67]. Kato and Launder proposed to replace the original production term of turbulent kinetic energy $P = \mu_t S^2$ with the term $P = \mu_t S \Omega$ where $S = \sqrt{2S_{ij}S_{ij}}$, $\Omega = \sqrt{2Q_{ij}Q_{ij}}$ and $S_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)$, $Q_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} - \frac{\partial u_j}{\partial x_i} \right)$, in an effort to eliminate the excessive generation of turbulence energy in the vicinity of stagnation points. This change has no effect in simple shear flows where $\Omega = S$. Near stagnation points, however, the flow is nearly irrotational and thus Ω is small which leads to the decrease of the turbulent kinetic energy. This modification led to improved results in flows around bluff bodies [67,68] and has been used again in deflagration simulations [69].

In Fig. 7 (left) the overpressure time series are presented for the new deflagration model and compared with the modified one (Section Main modification of the combustion model) and with the experiment (the models used are summarized in Table

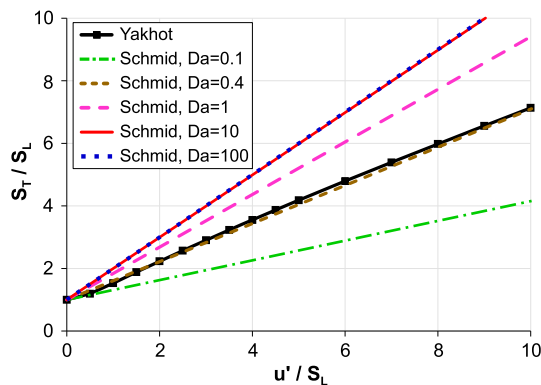


Fig. 6 – Comparison of Yakhot's [49] and Schmid's [64] relations for turbulent burning velocity.

1). We observe that the new model achieves very good agreement with the experiment. At the initial stage of deflagration when the flame is inside the enclosure the agreement is excellent in both internal and external overpressure. When the flame reaches sensor P09 (outside the vent) a sudden increase in external pressure occurs (around 0.077 s) which is overpredicted by the simulation. At the same time the pressure inside the enclosure stabilises around 5.8 kPa, similarly to the experiment. The time when pressure increases inside and outside the enclosure due to external explosion is also predicted very accurately. The rate of pressure increase inside the enclosure is captured very well. However, the maximum overpressure is overpredicted by approximately 17%. The fact that no heat losses were considered in the simulations (adiabatic walls and no radiation) may contribute to the overprediction of the maximum overpressure. Radiative and convective from the enclosure walls heat flux can affect the maximum overpressure as for example is shown in Refs. [11,70].

In the right part of Fig. 7 the combustion related parameters are presented for the flame inside or outside the enclosure for the modified and the new model. We observe that the new model has a small impact on the combustion inside the enclosure. On the other hand, outside the enclosure where turbulence is high and external explosion occurs, higher turbulent burning velocity values are predicted from the new model compared to the modified one which led to the better overpressure prediction.

The new deflagration model predicts also very satisfactory the flame position. In Fig. 8 the axial flow position (position of the tip of the hydrogen cloud that is pushed outside the vent) and flame position are presented for the experiment and the simulation. The position of the vent and the time when the flame reaches the vent are also shown with dashed lines. Regarding the axial flow position the agreement is initially excellent. At the later stages, a small overprediction is observed, approximately equal to 4–6 cm. The slope of the experimental curve is predicted accurately enough which indicates a good representation of the speed of the hydrogen cloud coming out from the vent. The qualitative agreement of the flame position is also very good. At the first stages of deflagration (until approximately 0.07 s) a small delay in the flame position is predicted from the simulation. As the flame approaches and passes through the vent the agreement becomes much better. However, after 0.078 s the predicted flame position deviates again from the experiment overestimating the measurements.

Due to strong coupling between combustion rate and turbulence characteristics, grid independency is often difficult to be achieved in deflagration modelling (e.g. Ref. [8,9,71]). In Fig. 9 the grid independency study is presented for the new deflagration model. We observe that grid resolution does not have a significant influence in the overpressure curves. The maximum overpressure is affected but not to a great extent. The maximum overpressure using the fine grid lies between the values achieved with the coarse and the medium one indicating a satisfactory grid convergence.

Analysis of the experiment using the new model

In Fig. 10 (left) the hydrogen mass fraction isosurface of 0.001 kg/kg along with the flame front surface (represented by

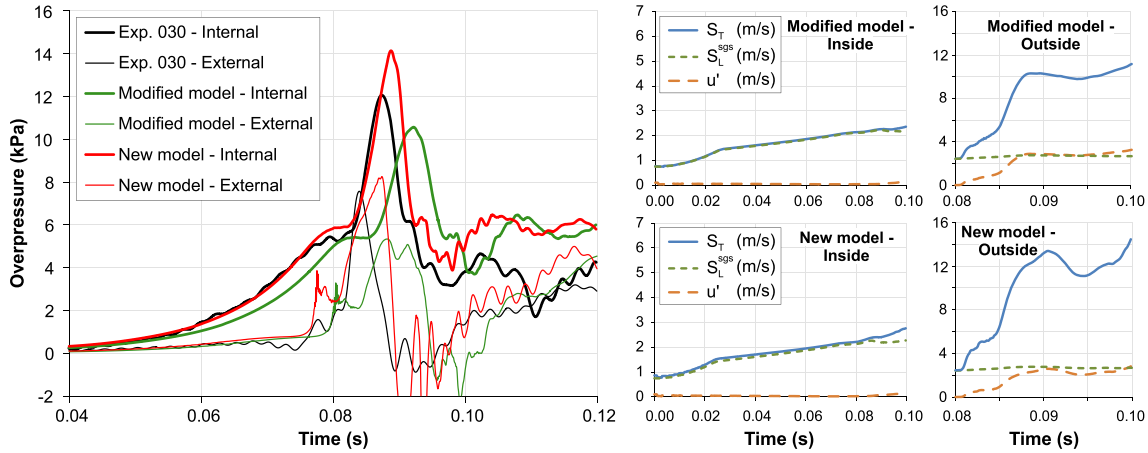


Fig. 7 – Results of the new model. Left: Overpressure inside and outside the enclosure. Right: Average combustion related variables at the flame front inside or outside the enclosure for the modified model (top) and the new (bottom) model.

Table 1 – Summary of the main characteristics of the three CFD deflagration models examined in the present study.

	Turbulence	Combustion
Initial model	k-ε/LES	Modified Yakhot's equation $S_T = \Xi_k \cdot \Xi_{lp} \cdot S_L \cdot \exp\left(\frac{u'}{S_T}\right)^2$
Modified model	k-ε	Based on Yakhot's equation $S'_T = S_L \cdot \exp\left(\frac{u'}{S'_T}\right)^2$ $S_T = \Xi_k \cdot \Xi_{lp} \cdot S'_T$
New model	Kato-Launder k-ε	Based on Schmid's equation $S'_T = S_L + u'(1 + Da^{-2})^{-1/4}$ $S_T = \Xi_k \cdot \Xi_{lp} \cdot S'_T$

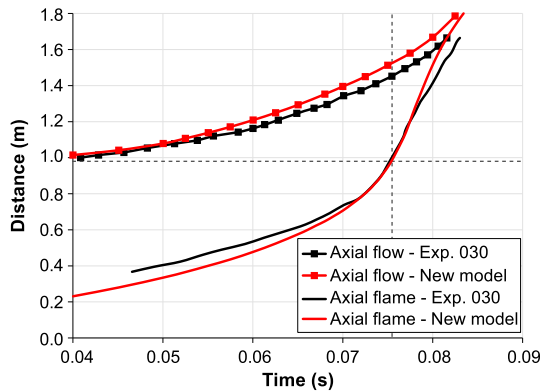


Fig. 8 – Axial flow and flame positions in experiment and simulation. Dashed lines indicate the position of the vent (0.98 m) and the time when the flame reaches the vent (0.0755 s).

water mass fraction isosurface of 0.0675 kg/kg - equivalent to progress variable of 0.5) are presented at various times of the external explosion. In the same diagrams the contours of u' are shown at the central xz plane along with the velocity vectors. Velocities reach high values which locally are larger than 150 m/s. The first picture corresponds to the onset of the external pressure sudden increase (0.081 s). We observe that

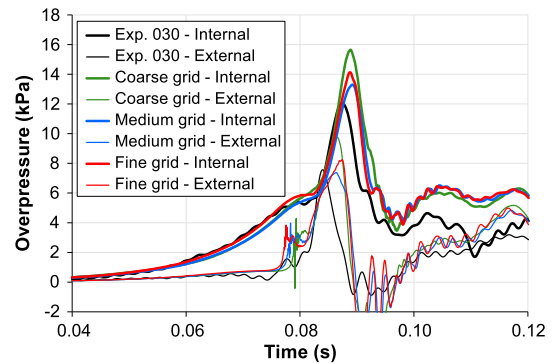


Fig. 9 – Grid independency study – Overpressure time series inside and outside the enclosure for different grid resolutions using the new model.

at this time the flame enters the area of high turbulence which has been formed lateral of the vent edges. Thus, the reaction rate increases and the pressure rises. At the following times (0.083, 0.085, 0.086 s) the external pressure continues to increase as the flame continues to propagate in the highly turbulent region. In the same figure (at the right) photos of the experiment are presented where both the hydrogen cloud which has been pushed outside the vent and the flame can be seen. The agreement with the experiment regarding the general shape and position is very satisfactory.

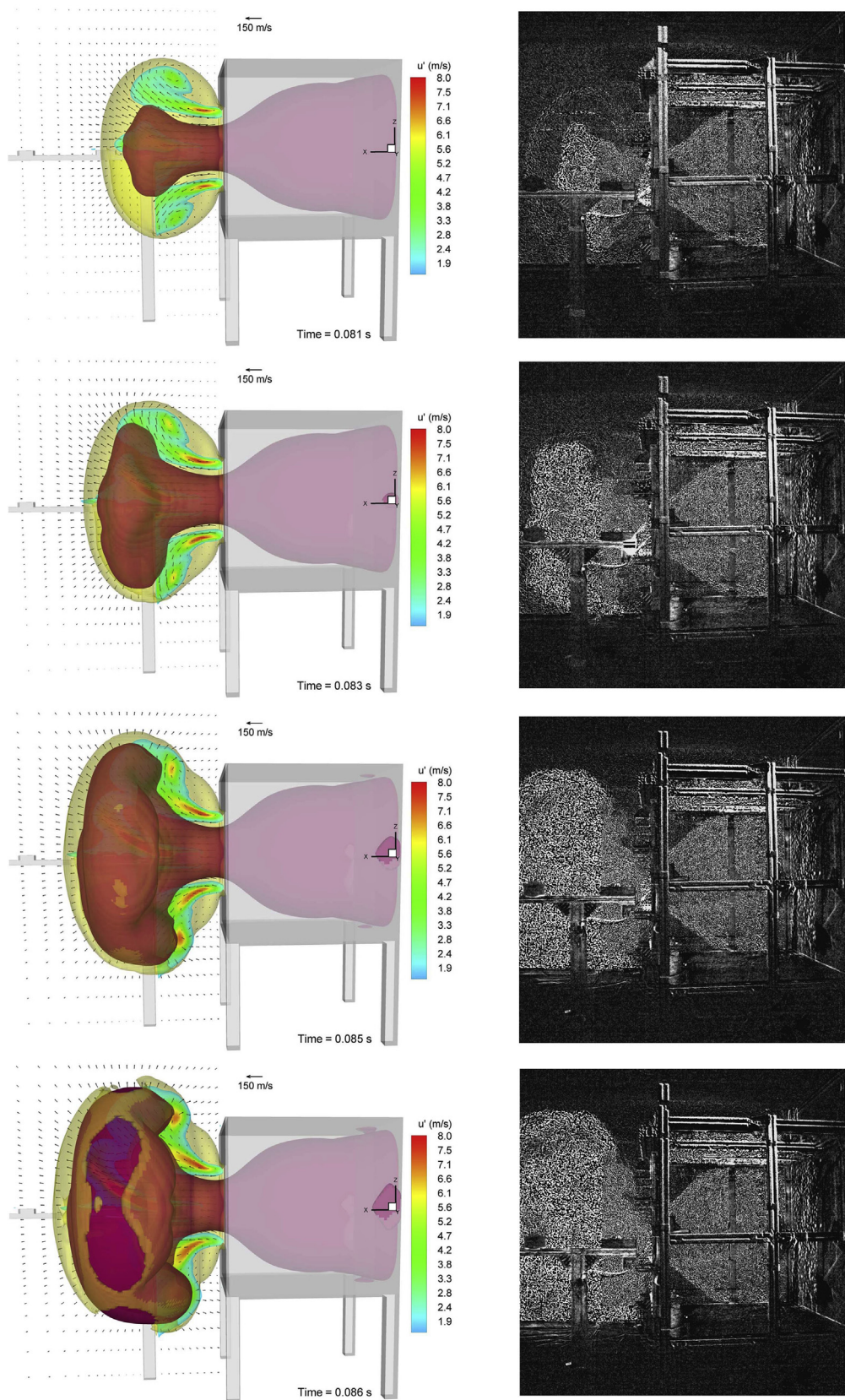


Fig. 10 – Left: Hydrogen mass fraction isosurface (yellow) of 0.001 kg/kg and flame front surface (dark red) along with u' contours (cutoff below 1.5 m/s) and velocity vectors (every 3 cells) at the central xz plane, at the time 0.081 s (onset of external pressure sudden increase), 0.083 s, 0.085 s and 0.086 s. Right: Photos of the experiment at the same times. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

The above observations are in agreement with the recent small scale detailed experiments of McGarry et al. (2017) [72] in which a premixed flame was investigated passing upon an obstacle in a channel. Even if the configuration of the experiment is different, the flow has similar features with our case at the moment when the flame passes through the vent. In McGarry et al. experiment turbulence had been generated behind the obstacle due to the imposed flow before the flame reached the obstacle. The flame did not transit to fully turbulent regime immediately after passing the obstacles but only when it was drawn into the recirculation region, exactly like in our case.

Another interesting observation from the simulation results in Fig. 10 is the fact that the highly turbulent region mainly develops inside the hydrogen cloud which has been formed outside the vent. Hydrogen starts to escape from the enclosure from the first stages of the explosion whereas turbulence mainly grows at the later stages when velocities values are high enough. As a result, the turbulent region develops inside the cloud. This may explain the fact that sharp borders of the cloud outside the vent and low dilution with the surrounding air are observed in the experiments of Proust and Leprette (2010) [23] and Daubech et al. (2017) [73] (in these experiments the flammable mixture was seeded with particles to make the cloud visible).

Pressure increase mechanism

The pressure increase inside the enclosure is a result of the competition between two phenomena [59,74]: The increase in volume that combustion imposes due to heat generation and the decrease of volume that the vent impose due to venting of the flow. The venting rate $\dot{V}_{vent}$ can be easily estimated through the relation $\dot{V}_{vent} = \iint_{S_{vent}} u_n ds$ where S_{vent} is the vent area

and u_n the normal to the vent velocity component. Following Quillatre et al. (2013) [59], the rate of volume increase due to combustion, $\dot{V}_{comb}$, is estimated as the difference between the rates of volume generation of the burned area minus the rate of consumption of the unburned one. The volume in the unburned region is linked to that of the burned region through the relation $V_u = V_b/E$, where $E = \rho_u/\rho_b$ the expansion ratio and ρ_u , ρ_b the densities of the unburned and burned regions respectively. Thus,

$$\dot{V}_{comb} = \frac{\partial V_b}{\partial t} - \frac{\partial V_u}{\partial t} = \frac{\partial V_b}{\partial t} - \frac{1}{E} \frac{\partial V_b}{\partial t} = \left(1 - \frac{1}{E}\right) \frac{\dot{\omega}_{H_2O}}{\rho_b Y_{H_2O}^b} \quad (6)$$

where $Y_{H_2O}^b$ is the mass fraction of water in the burned region. In this relation, the rate of increase of the volume of the burned region, $\partial V_b/\partial t$, was estimated through the mean production rate of products (water in our case), $\dot{\omega}_{H_2O}$ (kg/s), which is determined by the combustion model. The total $\dot{V}_{comb}$ is estimated by adding the contributions of every computational cell inside the enclosure.

The mechanism which raises the pressure is demonstrated in Fig. 11 in which the venting rates are presented as functions of time along with the pressure inside the enclosure. We observe that when $\dot{V}_{comb} > \dot{V}_{vent}$ pressure increases because combustion overcomes venting whereas when $\dot{V}_{comb} < \dot{V}_{vent}$ the pressure decreases because combustion is not strong enough to maintain the pressure. In the same diagram the

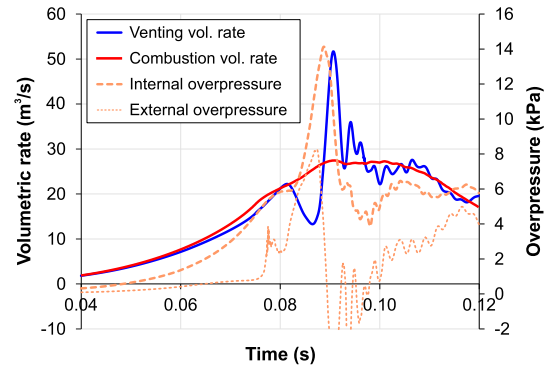


Fig. 11 – Volumetric flow rate through the vent along with the volumetric rate generated by combustion inside the enclosure. Internal and external overpressures are also shown.

pressure outside the vent is also presented in order to demonstrate the impact that the pressure outside the vent has on the venting rate: its increase leads to the decrease of the venting rate. This is normal as the increase of the pressure outside the vent lead to decrease of the pressure difference at the two sides of the vent and thus to the decrease of the flow rate. On the other hand, the combustion rate inside the enclosure is independent of the external pressure.

Borghi diagram

Turbulent flames can be categorized according to ratios of turbulent scales to flame scales. Many diagrams have been proposed for that purpose, which are known as Borghi diagrams [53,75]. A type of Borghi diagram is presented in Fig. 12. The ratio of turbulent length scale (estimated through the relation $k^{3/2}/\epsilon$) to laminar flame thickness is on the horizontal axis and the ratio of u' to laminar flame speed on the vertical axis. The points noted in the diagrams have been estimated in front of the flame tip at the upper part of the flame which rolls up entering the high turbulent zone (see Fig. 10). The values of the variables at these points are presented in Table 2. We observe that all points lie inside the reaction sheet region (also known as thickened wrinkled flames or thin reaction zone [76]) being

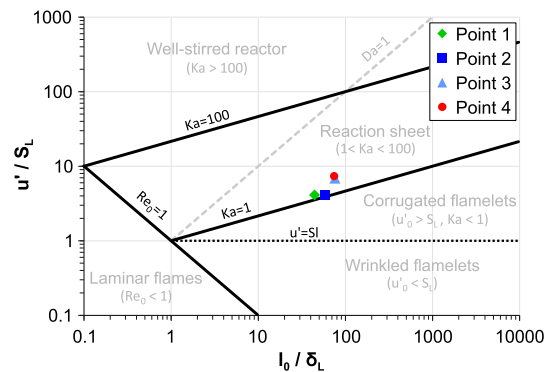


Fig. 12 – Borghi diagram. The points have been estimated in front of the flame tip at the upper part of the flame which rolls up entering the high turbulent zone (Fig. 10). Variables values are presented in Table 2.

Table 2 – Parameters of the points presented in Borghi diagram (Fig. 12). The values of $\delta_L = 0.37 \cdot 10^{-3}$ m and $S_L = 0.64$ m/s were used for all points.

Point	Time (s)	k (m ² /s ²)	ε (m ² /s ³)	u' (m/s)	l_0 (m)	u'/S_L	l_0/δ_L
1	0.081	10.35	2031.8	2.63	0.0164	4.1	44.3
2	0.083	10.63	1599.1	2.66	0.0217	4.2	58.6
3	0.085	29.50	5697.8	4.43	0.0281	6.9	76.0
4	0.086	33.32	6995.0	4.71	0.0275	7.4	74.3

very near the corrugated flamelets region. In reaction sheet region, turbulence can disturb the flame preheat zone but cannot disturb the reaction zone which retain the laminar characteristics whereas in corrugated flamelets region turbulence cause intense flame wrinkling without modifying the internal structure of the flame. The area in Borghi diagram where the points lie agrees with the main assumption of the deflagration model that burning velocity increases due to flame wrinkling.

We should note that when the flame is inside the enclosure, the flame is laid inside the laminar regime of the Borghi diagram. Borghi diagrams should be used with caution and only for qualitative analysis due to the assumptions made [76]. This is particularly true for hydrogen because thermo-diffusion instability can change the limits in Borghi diagram as for example was demonstrated in Ref. [77]. Besides the uncertainties, this diagram is presented here in order to have an overview of the turbulent regime and to be used as a reference for comparison with other deflagration cases. For example it is interesting to note that in McGarry et al. (2017) [72] experiments the turbulent flame lied also mainly inside the reaction sheet regime and near the corrugated flamelets region.

Discussion

From the analysis of this work a formulation of the combustion models which rely on turbulent burning velocity concept can be made. The important components for a successful CFD deflagration model are three:

- A submodel for the effect of flame instabilities
- A submodel for the effect of turbulence that exists in front of the flame front
- The connection of the above submodels

In the present model, $\Xi_k \cdot \Xi_{lp}$ factor was used as submodel for the effect of flame instabilities. This is a simple algebraic model which have been used successfully in the past (e.g. Ref. [28,78]). More complicated models, like the model of Bauwens et al. (2018) [79], have recently been developed and could be studied as flame instabilities submodel in CFD simulations.

As submodel for turbulence effect on the flame front, the Schmid's and Yakhot's equations were used. Many relations for turbulent burning velocity exist in the literature and could also be tested, like for example Bray's [80] and Zimont's [81] relations that have been utilized in FLACS [82] and GASFLOW-MPI [83] CFD programs respectively. Validation against many experiments needs to be conducted in order to examine which relation is performed best. We should keep in mind, however, that

these relations are heavily relied on u' prediction and consequently on the utilized turbulence model. Validation would be more reliable if we knew that u' is predicted correctly. For that purpose, experiments that measure velocity fluctuations are needed. However, to the best of our knowledge, u' have been measured only in few small scale deflagration experiments like for example that in Ref. [84] and in Refs. [85,86].

As far as the connection of the submodels is concerned, the modelled phenomena were considered to be independent from each other and thus to wrinkle the flame at different scales. As a result their interaction was modelled through the multiplication of the $\Xi_k \cdot \Xi_{lp}$ factor with the S'_T . A similar assumption is made in the model of Bauwens et al. (2011) [24]. However, the validity of this assumption is questioned. As a result different interaction of the modelled phenomena could be considered. For example in FLACS the maximum of a "quasi-laminar burning velocity" (which is used to account for flame instabilities) and the turbulent burning velocity is used [82]. Thus, they consider that the effects of turbulence and flame instabilities are similar and cannot add up. A similar approach is also used in the models of Refs. [87,88]. The geometry of the problem plays a crucial role because one phenomenon may dominate over the other. For example, in a free unconfined deflagration flame instabilities will dominate whereas in high congested domain turbulence will dominate. In vented deflagrations cases with no obstacles inside the enclosure, a submodel for flame instabilities is indispensable in order to predict the correct flame propagation as the flame propagates inside the enclosure. For example, one of the simulations conducted in Ref. [8] in which artificial high level of initial turbulence was used to trigger turbulent combustion resulted in poor agreement with the experiment.

An important question is to what extent Rayleigh-Taylor mechanisms due to density difference between the burned and unburned area that is described elsewhere (e.g. Ref. [18,24,31,89]) influences the combustion rate and whether or not special explicit models are needed to model it. Two models have appeared in the literature, the one of Bauwens et al. (2011) [24] and the one of Keenan et al. (2014) [31], in an effort to improve the results of their CFD simulations which underpredicted the external overpressure. However, in the present work, in which the effect of external explosion is evident in the experiment, no special modelling of the Rayleigh Taylor instability needed to be made in order to capture the correct overpressure. The model used was based only on the flame front instabilities which develop at the initial stage of the explosion and on the turbulence that exists in front of the flame front. The model was able to capture the overpressure curve very accurately. The sudden increase of the reaction rate occurred only when the flame entered the highly turbulent area (which had been formed before the flame reached that area) outside the vent. Moreover, the flame shape and position are in very good agreement with the experiment. Consequently, it seems that the Rayleigh-Taylor instability does not influence the phenomenon here.

A common feature of the models of Bauwens et al. [24] and Keenan et al. [31] is that both have been used in the context of LES. A question that rises is whether LES method was the reason why external explosion could not be predicted accurately without the use of special submodels. LES is considered

superior to the URANS method and it is expected that LES could predict the turbulence generated outside the vent and thus external explosion better than URANS models. One reason for the poor performance of the LES models could be the very low levels of resolved turbulence. This can be caused by a number of factors such as: inappropriate numerical schemes, poor grid resolution and lack of initial perturbations to trigger numerically the turbulence. Any LES analysis should include information about the resolved part of turbulence and ensure that velocity fluctuations, which wrinkle the flame, exist in order LES to be meaningful (as for example in Ref. [59,60,90,91]). Unfortunately, this is not always done. Another reason could be the use of combustion models which have not been developed specifically in the context of LES. LES oriented combustion models are for example those in Ref. [61,63,92] which have been used with success in deflagration simulations (e.g. in Ref. [91,93]). Therefore, the LES conducted in this work cannot be considered as a rigorous LES study too because we chose to use similar simulation set-up as the one used in previous studies (e.g. Ref. [5–8,28,29]) of the initial combustion model.

On the other hand, URANS models seem to be able to predict the external explosion effect. This is shown in the present work but it can also be concluded from the work in Ref. [8]. In Ref. [8] two LES and two URANS approaches were used in order to model the same experiment as that of Bauwens et al. [24] and Keenan et al. [31] works (FM-Global experiments). In LES approaches explicit modifications of the combustion models were required in order to capture the external explosion effect whereas in URANS approaches no special treatment was made. Even though LES achieved better agreement with the experiment, this should be attributed to the calibration of the external explosion model to match the experimental results. URANS models on the other hand did not achieve good quantitative agreement with the experiment but they managed to predict the sudden increase of the external pressure.

A global deflagration model, capable for accurate predictions of a wide variety of cases is very difficult to be developed due to the high number of uncertainties in the physics of the phenomenon. Besides combustion modelling, turbulence modelling is also very important mainly due to the strong coupling between turbulence and combustion. Even if the turbulent combustion rate was known, a proper turbulence model would be needed in order to predict with accuracy the flow dynamics. Consequently, extensive model validation needs to be conducted against a wide variety of well described accurate experiments. CFD studies with models which accounts for different mechanisms for flame acceleration should present how each mechanism is predicted to increase the reaction rate in order to identify which phenomenon dominates in each stage of the deflagration. This contributes also to a more reliable model validation.

Conclusions

The deflagration experiment of 18% v/v hydrogen-air homogenous mixture in a 1 m³ vented enclosure conducted by

KIT was studied. The main conclusions of the current work can be summarized in the following points:

- The new CFD deflagration model, which was implemented in the ADREA-HF code, improved the initial model to a great extent reproducing the experiment accurately enough. The new model utilizes a different coupling between the submodels for the flame-front instabilities and turbulence, a different relation to account for turbulence existing in front of the flame front and the Kato-Launder modification of the k-ε model.
- The results of the CFD simulations indicate that unresolved phenomena like flame instabilities that occur in the flame front are responsible for flame acceleration when the flame is inside the enclosure. Turbulence is of major importance for the external explosion event.
- The violent increase of the external pressure due to external explosion does not occur immediately after the flame has exited the vent but later, when the flame enters the turbulent recirculation zone.
- Rayleigh-Taylor instability does not seem to influence the phenomenon. No special modelling of the Rayleigh-Taylor instability needed to be made. Good agreement with the experiment was achieved by proper modelling of the turbulence existing outside the vent.
- More effort and research should be made for LES in medium and large scale deflagrations. At the moment URANS models seem easier to be applied and more effective.

As future work, the new model should be evaluated against other experiments in order to investigate its range of capabilities. Congested vented deflagration modelling has an additional challenge and more efforts should be paid as they represent better real case scenarios.

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Appendix

Here we prove that the modified model of Equations (3), (4) give higher S_T than the initial model of Equation (2). We will prove it for the general form of S_T , that is $S_T/S_L = f(u'/S_L)$.

We suppose constant values of S_L and u' and that $S_L^{sgs} = a \cdot S_L$ where $a > 1$. The initial model gives the following turbulent flame velocity:

$$\frac{S_T}{S_L^{sgs}} = f\left(\frac{u'}{S_L^{sgs}}\right) \Rightarrow S_T = f\left(\frac{u'}{a \cdot S_L}\right) \cdot a \cdot S_L \quad (7)$$

The modified model gives the following turbulent flame velocity:

$$\left. \begin{aligned} \frac{S'_T}{S_L} &= f\left(\frac{u'}{S_L}\right) \\ S'_T &= a \cdot S'_T \end{aligned} \right\} \Rightarrow S_T^{new} = f\left(\frac{u'}{S_L}\right) \cdot a \cdot S_L \quad (8)$$

It is obvious that if f is an increasing function (which is the case of turbulent flame velocity correlations at least at u' range of interest) then $f\left(\frac{u'}{S_L}\right) > f\left(\frac{u'}{a \cdot S_L}\right)$. Thus, $S_T^{new} > S_T$.

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